packages/graph-core/src/embedding-search.ts

Kind: added · Baseline: 8e0dc031dcfc · Target: NovaDiff · Generated: 2026-05-19T19:39:06.806Z

Overview

A new file `packages/graph-core/src/embedding-search.ts` adds a lightweight semantic search engine that operates on pre-computed vector embeddings of graph nodes.

Key changes

- **Imports** – `GraphNode` from `./types.js` and `SearchResult` from `./search.js` (lines 1-2).
- **`SemanticSearchOptions`** (lines 4-7) introduces optional `limit`, `threshold`, and `types`.
- **`cosineSimilarity`** (lines 14-30) computes dot product and magnitudes, returning `0` when either vector has zero magnitude.
- **`SemanticSearchEngine`** (lines 37-83) stores nodes and embeddings, and exposes `hasEmbeddings`, `addEmbedding`, `search`, and `updateNodes`.
- **Search logic** (lines 54-78) filters by node type, applies a similarity threshold, scores as `1 - similarity`, sorts ascending, and slices to the requested limit.

Impact

- Deterministic similarity scoring; zero-magnitude vectors return `0`, avoiding NaNs.
- Encapsulates search logic in a single class; the options interface centralizes configuration.
- Performs a linear scan over all nodes; suitable for small graphs but may need optimization for larger datasets.
- No changes to existing public APIs; the new module is isolated.

Risks & follow-ups

- No unit tests cover the new engine; run `npm test` to confirm no accidental breakage.
- Linear search could become a bottleneck; benchmark against the current search implementation.
- Embedding keys must match `node.id`; mismatches will silently skip nodes.
- Build verification: run `npm run build` to confirm the new file is included.